

AI-Powered Cloud Masking for Satellite Imagery

Project Report

Course: Remote Sensing and Satellite Imagery (Spring 2025)
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1. Introduction

This project report details the development and assessment of an AI-based system for automated cloud detection in multispectral satellite imagery.

The project aims to create binary cloud masks by applying machine learning & deep learning techniques. We start by performing EDA, followed by the selection and optimization of a deep learning model, balancing accuracy and efficiency. Several deep learning architectures— U-Net, U-Net++, DCNet, DeeplabV3(+), SegFormer and Cloudformer—were tested through multiple experiments. These trials involved investigating different concepts, including dilated convolutions, spatial pyramid structure and other state of the art concepts for segmentation models.

This report details the complete project pipeline, starting with a review of existing literature and papers which our work was based on (Section 2) and an analysis of the provided dataset (Sections 3 & 4). It then describes the data preprocessing steps (Section 5), the selection and training of machine learning & deep learning models (Section 6), and a thorough analysis of their performance (Section 7). potential future work (Section 8), final conclusions (Section 9), team contributions (Section 10), and references (Section 11) are also presented.

2. Literature Review

A thorough review of existing literature was conducted to understand established and

state-of-the-art approaches for semantic segmentation, particularly in the context of cloud masking from satellite imagery. This informed our exploration of various model architectures. We investigated several well-known architectures as potential foundations for our solution:

- **U-Net:** A widely used CNN for image segmentation, employs a symmetric encoder-decoder architecture. The encoder down-samples for context, and the decoder up-samples for localization. Skip connections between corresponding encoder and decoder layers facilitate the reuse of fine-grained features.
- **U-Net++:** An extension of U-Net that aims to bridge the semantic gap between encoder and decoder feature maps. It introduces nested and dense skip connections, creating a more flexible feature fusion scheme that integrates features at different semantic scales, potentially improving segmentation accuracy, especially for objects of varying sizes.
- **DCNet (Deformable Convolutional Network):** This cloud masking network uses an encoder-decoder architecture with Deformable Convolution Blocks (DCBs) to handle variable cloud geometry. DCBs learn offsets for convolutional filters, enabling better adaptation to irregular cloud shapes compared to standard convolutions. Skip connections combine spatial details and semantic features in the decoder.
- **DeeplabV3(+):** A family of models known for employing atrous (dilated) convolutions within the encoder. This allows the network to capture multi-scale contextual information by probing features at different resolutions without increasing parameters or computation significantly. DeeplabV3 utilizes the Atrous Spatial Pyramid Pooling (ASPP) module to aggregate context at multiple scales. DeeplabV3+ further refines segmentation boundaries by adding a simple yet effective decoder module.
- **SegFormer:** A relatively recent architecture based on the Transformer model. SegFormer uses a hierarchical Transformer encoder that outputs multi-scale features without needing positional encodings, making it adaptable to different input sizes. It combines this with a lightweight All-MLP (Multilayer Perceptron) decoder, resulting in an efficient and powerful segmentation model.
- **Cloudformer:** This cloud detection framework uses a CNN and Transformer. Its

Spatial Pyramid Structure (SPS) Encoder with convolutions extracts multi-scale features. The decoder includes a CNN-based Supplementary Aggregation (SupA) Decoder and a Transformer module that processes queries to predict class and mask embeddings. The final segmentation combines these predictions with pixel-level features and incorporates asynchronous position encoding for the Transformer.

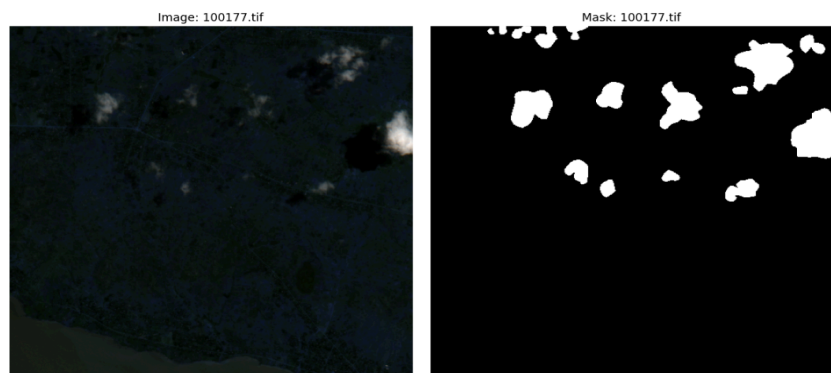
The choice of these models was based on [this paper](#) which provided this table which we used as our initial reference before performing in depth trials with different models:

Model	IoU	OA	Precision	F1	Recall
UNet	0.7498	0.9354	0.8116	0.8570	0.9079
CDNet	0.7472	0.9325	0.7876	0.8553	0.9357
DANet	0.7214	0.9219	0.7510	0.8382	0.9482
ResNet	0.6332	0.8819	0.6522	0.7754	0.9561
DeeplabV3	0.8063	0.9516	0.8467	0.8928	0.9442
DeeplabV3Plus	0.8115	0.9539	0.8628	0.8960	0.9318

This paper encouraged us to use DeeplabV3+ as it was more consistent across different metrics and provided the best F1-Score which agreed with our analysis at the very end.

3. Dataset Description

The dataset provided for this project consists of approximately 10,000 (512*512 px) satellite image pairs, each pair consists of a tiff image file consisting of 4 bands (R,G,B ,NIR) and a binary mask highlighting the segmented clouds with 1s for the cloud mask and 0s for the background making it basically a binary classification task for each pixel.

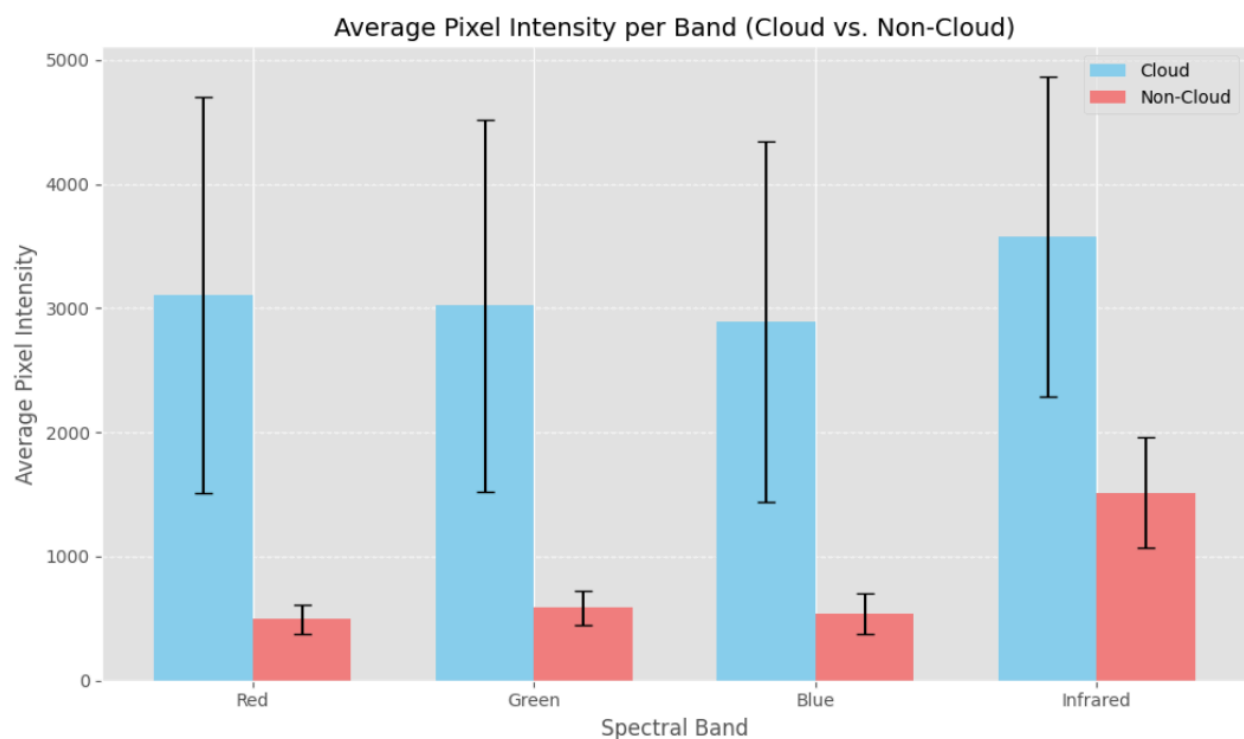


We divide this dataset with a ratio of 0.9:0.1 train-test split and perform 5-fold cross validation on the train part.

4. Exploratory Data Analysis (EDA)

To gain a better understanding of the dataset characteristics, we performed Exploratory Data Analysis (EDA) focusing on the distribution of cloud cover within the provided ground truth masks. Key steps and findings include:

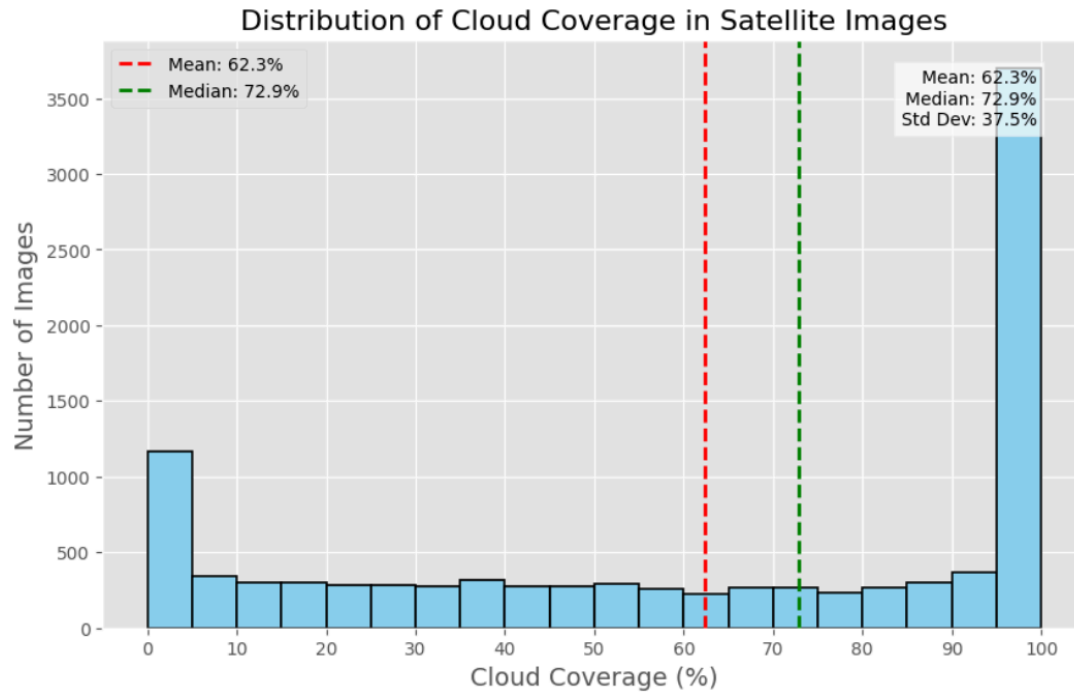
- **Band Analysis:** In this step we analyze each spectral band (Red, Green, Blue, Near-Infrared) by calculating mean, standard deviation, min, and max pixel values. Generate histograms to visualize intensity distribution and identify anomalies. This per-band analysis informs preprocessing and model development.



This shows that cloud pixels are generally brighter than background pixels (duh!) and that the IR band has a slightly different distribution than the other bands making it a bit more valuable considering that it adds another dimension of information to our model.

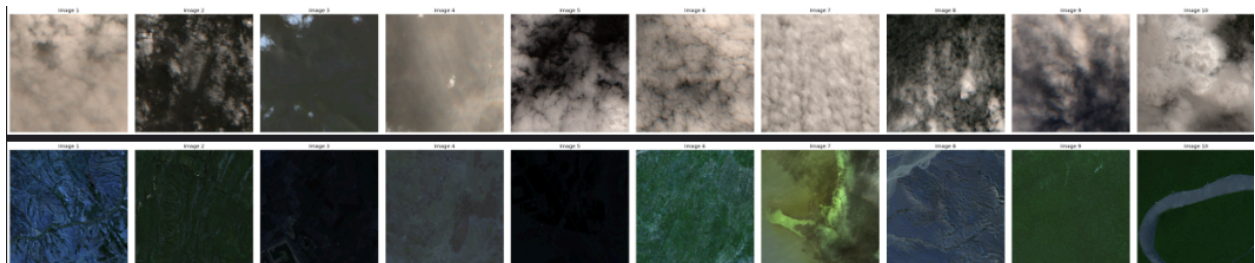
- **Cloud Coverage Calculation:** For each mask file, we calculated the cloud coverage percentage by dividing the number of cloud pixels (identified by a pixel value of 1) by the total number of pixels in the mask.
 - **Image Categorization:** Based on the calculated coverage, we categorized the images into three groups:
 - **fully_clouded:** Images with more than 80% cloud coverage.

- partially_clouded: Images with cloud coverage between 20% and 80%.
- cloud_free: Images with 20% or less cloud coverage.



This shows that the data is biased and needs stratification as the number of fully clouded images almost represents more than 45% of the data showing imbalance in the distribution.

- **Extreme Case Identification:** We specifically identified images that were either completely covered by clouds (all mask pixels = 1) or completely free of clouds (all mask pixels = 0).
 - Number of all-cloud masks: 2713 (25.66%)
 - Number of no-cloud masks: 701 (6.63%)



The images in the top row show fully clouded images while the second shows images with no clouds.

This was when we noticed that there's noise in the given data, by observing the third image in the first row we can see that it's incorrectly classified as fully clouded with all 1s mask (We will dive deeper into the noisy data handling in the preprocessing module).

5. Preprocessing Module

To ensure the satellite imagery and corresponding masks were optimally prepared for model training and evaluation, a series of preprocessing steps were implemented.

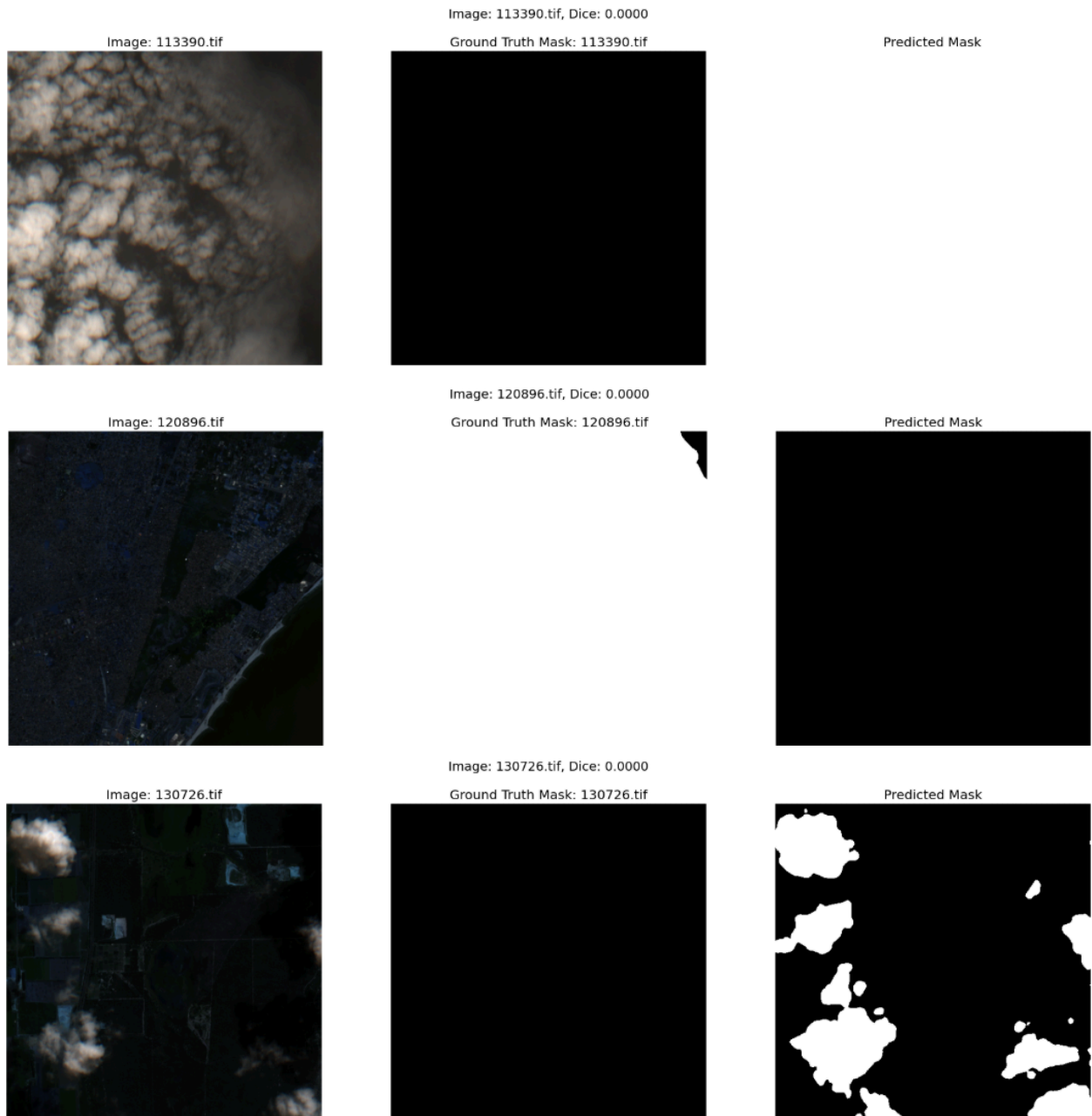
- **Data Augmentation:** To mitigate overfitting and improve the model's ability to generalize to unseen data exhibiting natural variations, several data augmentation techniques were applied during training:
 - **H/V Flipping & Rotations:** These transformations were applied with a certain probability to make the model invariant to the orientation of clouds within the image frame simulating different orientations by the satellite.
 - **Random Brightness and Contrast Adjustment:** Minor random adjustments to brightness and contrast were introduced to help the model become robust to variations in illumination conditions inherent in satellite imagery acquisition but i found that this potentially hurt the model so i commented it as the model was sensitive towards brightness changes.
- **Band Selection:** All four spectral bands provided in the dataset – Red (R), Green (G), Blue (B), and Near-Infrared (NIR) – were utilized as input to the models. The decision to include all bands was based on the understanding that each band potentially carries unique spectral information useful for discriminating clouds from other features.
- **Other Preprocessing:** Just basic normalization dividing by 2^{16} and binarization for the masks which wasn't really needed.

Each of these preprocessing choices was made to enhance the quality and diversity of the training data, ultimately aiming to improve the accuracy and robustness of our cloud masking model. Normalization helps in stable training, augmentation prevents overfitting, the inclusion of the NIR band provides valuable spectral information, and addressing noisy data ensures the model learns from accurate labels.

Now to address the elephant in the room which was that ~8% of the data as per our estimations was corrupted where some data points were sometimes the complete inverse of the correct mask, for instance an image which was fully cloudy would have a mask label with all 0s and vice versa.

These noisy data points were catastrophic, reducing the models ability to learn and discover features towards achieving its goal of cloud segmentation.

These are samples for these data points with their different types:



Our preprocessing approach to solve this major problem started by manually collecting the

indices of those noisy masks and excluding them from the training process, then similar to semi-supervised techniques we used the trained model to relabel those mislabelled data points to use them in our training process, this process was performed for multiple iterations (~8 times) till the model converged and the results were decent as shown in the predicted mask in the above figures (those masks labelled as predicted mask were used instead of the ground truth to train our model on data we are positive that they are correct to avoid the destructive nature of training our model on mislabelled data points) and hence this was the breakthrough in our approach which really made the major difference in the training process.

6. Model Selection & Training Module

Our approach involved exploring both classical machine learning techniques and a range of state-of-the-art deep learning architectures for semantic segmentation to determine the most effective method for cloud masking in the given satellite imagery.

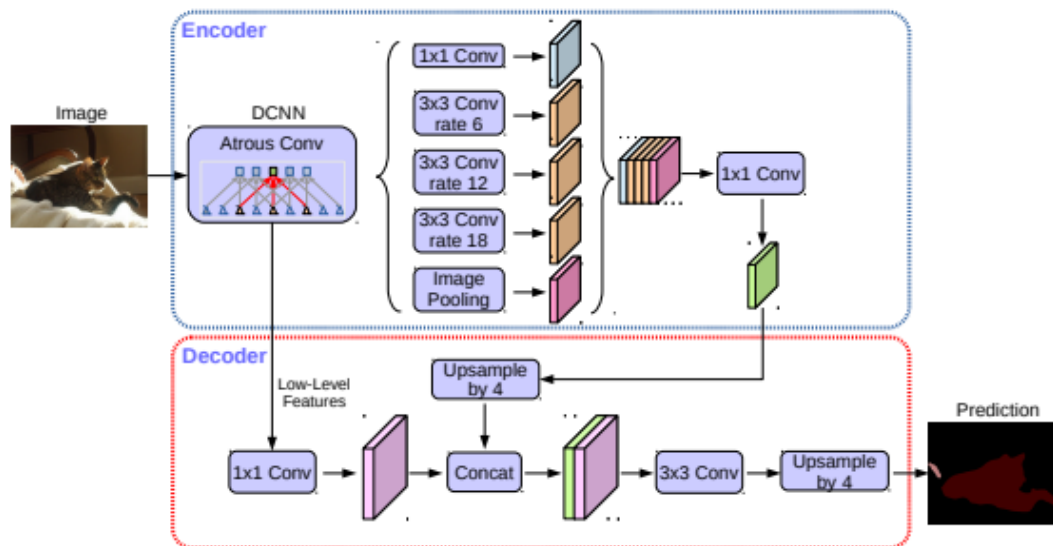
- **Classical Model:** We used SVM as our classical model trained on the pixel intensity values from the different bands, this was a basic approach which performed surprisingly well achieving 0.77 dice score which was very impressive as it was trained only on a subset of the training set and without relabelling or even cleaning the noisy data points, i expect it would easily reach ~0.85 if we provided it with the whole pipeline with cleaning the data and tuning the hyperparameters.
- **Deep Learning Model:** Guided by our literature review and the goal of achieving high accuracy and efficiency, we investigated several deep learning architectures known for their strong performance in image segmentation tasks:
 - U-Net & U-Net++
 - DCNet
 - DeeplabV3(+)
 - SegFormer (B2-B3-B4)
 - Cloudformer

These models represent diverse approaches, including standard encoder-decoders (U-Net), architectures with enhanced skip connections (U-Net++), deformable convolutions (DCNet), atrous convolutions with spatial pyramid pooling (DeeplabV3+), and Transformer-based methods (SegFormer, Cloudformer). Initial experiments and comparisons were conducted across these architectures (details available in the Appendix).

Final Deep Learning Model Selection: DeeplabV3+

Based on our comparative experiments, considering both segmentation performance (Dice coefficient) and computational efficiency (training time, model size), **DeeplabV3+** emerged as the most promising and well-balanced architecture for this specific cloud masking task.

DeeplabV3+ Architecture:



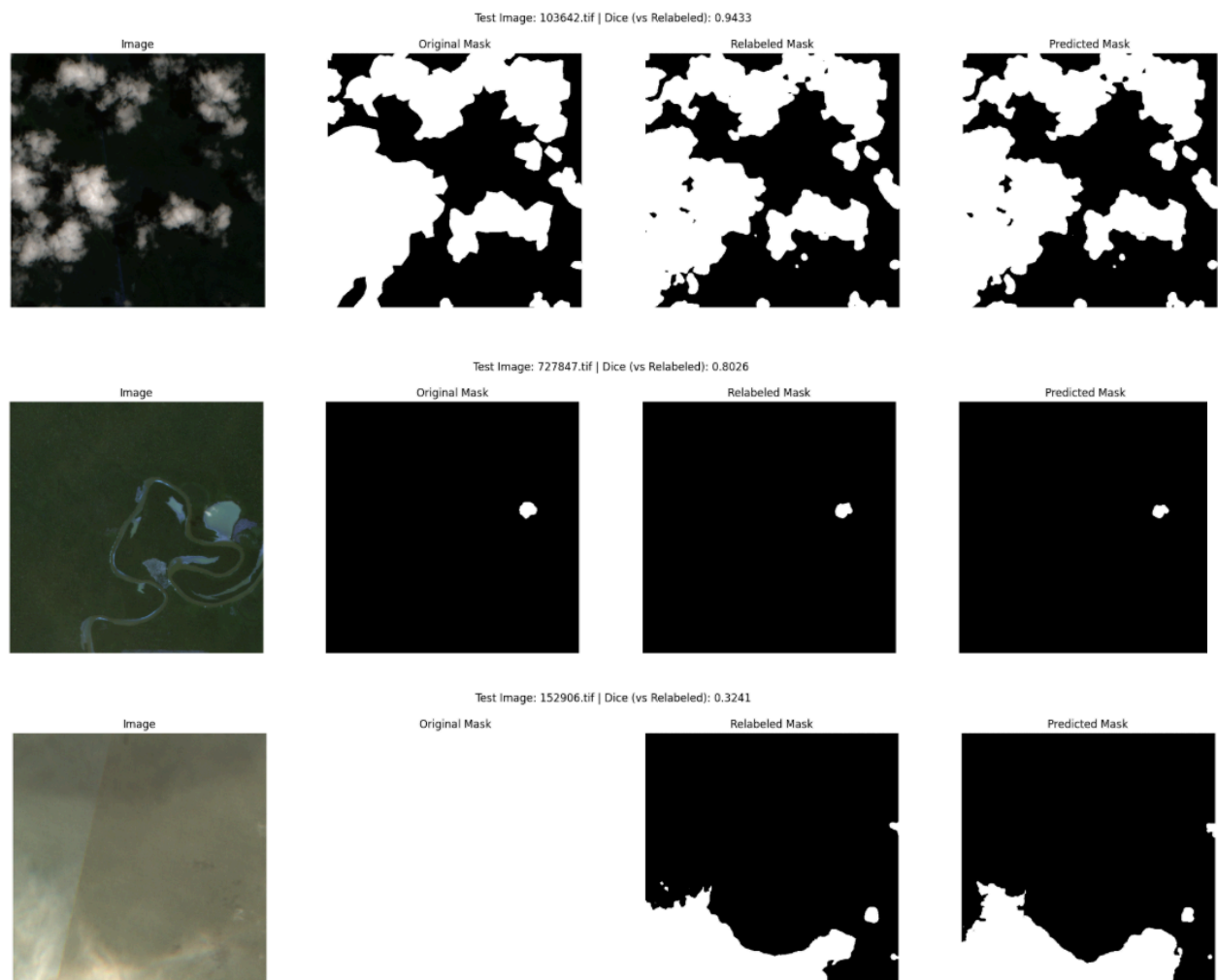
- **Encoder:** DeeplabV3+ typically employs a powerful Convolutional Neural Network (CNN) backbone, often a modified ResNet or Xception network, pre-trained on a large dataset like ImageNet, as its feature extractor (encoder). This encoder uses *atrous (dilated) convolutions*, which allow the network to capture features at multiple scales without significantly increasing the number of parameters or losing resolution rapidly.
- **Atrous Spatial Pyramid Pooling (ASPP):** A key component is the ASPP module, applied at the end of the encoder. ASPP uses parallel atrous convolutional layers with different dilation rates, effectively probing the incoming features at multiple scales. It also includes image-level features via global average pooling. This allows the model to capture contextual information from varying receptive fields, crucial for segmenting objects (like clouds) of different sizes.
- **Decoder:** Unlike DeeplabV3, DeeplabV3+ incorporates a simple yet effective decoder module. This decoder upsamples the output features from the ASPP module and concatenates them with corresponding low-level features from earlier in the encoder backbone (typically features with higher spatial resolution). This fusion helps to refine the segmentation boundaries, leading

to more precise masks, which is particularly important for accurately segmenting complex cloud edges.

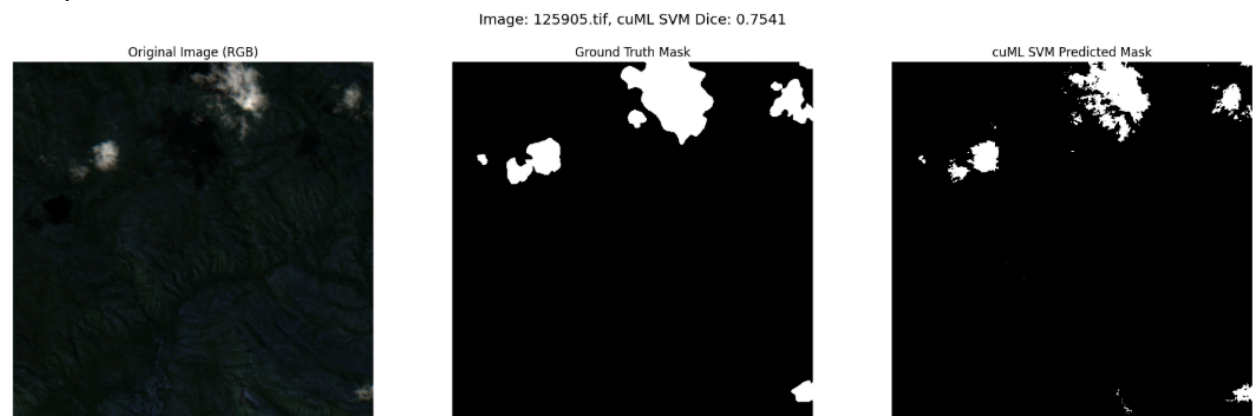
We developed and trained a DeeplabV3+ model using PyTorch, employing a 5-fold cross-validation strategy on the 4-band satellite imagery to ensure robust evaluation. A significant part of our approach involved a custom data loader designed to handle label noise; it incorporated relabeled masks for known noisy samples within the training folds, while validation consistently used the original masks to assess generalization to unmodified ground truth. The training process for each fold ran for approximately 10-15 epochs with a batch of size 10 & a learning rate of $5e-4$, utilizing the Adam optimizer, a combined Binary Cross-Entropy (BCE) and Dice loss function tailored for segmentation, and gradient clipping techniques to maintain stability.

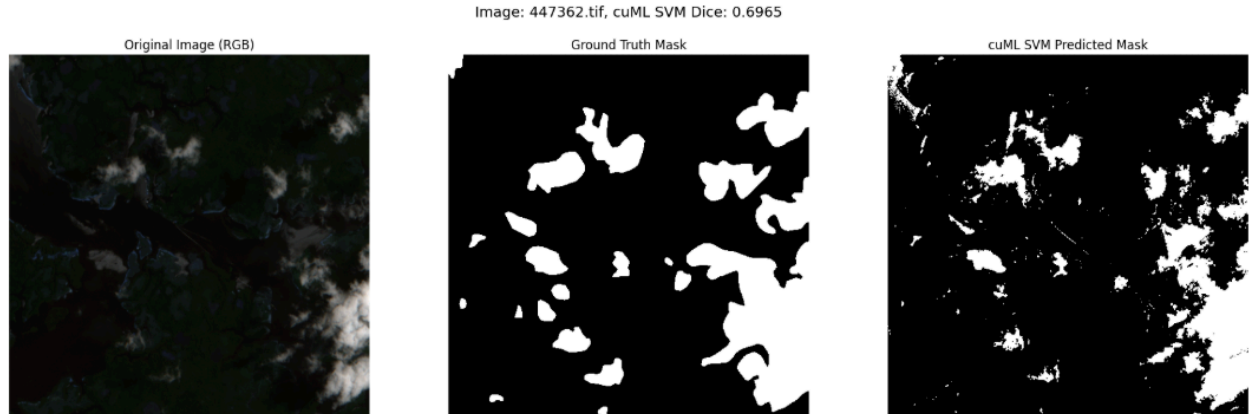
Impressively, when evaluated on the final hidden test set – which notably also included noisy data points similar to those encountered in the original training data – the model achieved a Dice coefficient of **0.8966**. This strong performance on potentially imperfect ground truth underscores the model's effective generalization. To further quantify the impact of label noise on evaluation, the model's performance significantly increased to approximately **~0.94** when assessed on a cleaned version of the test set from which these noisy data points were removed, highlighting its high accuracy on reliable data (though these noisy points were classified as noisy by the model itself which isn't good as this compromises the integrity of the test set).

Sample results from the model:



Sample results from the SVM model:





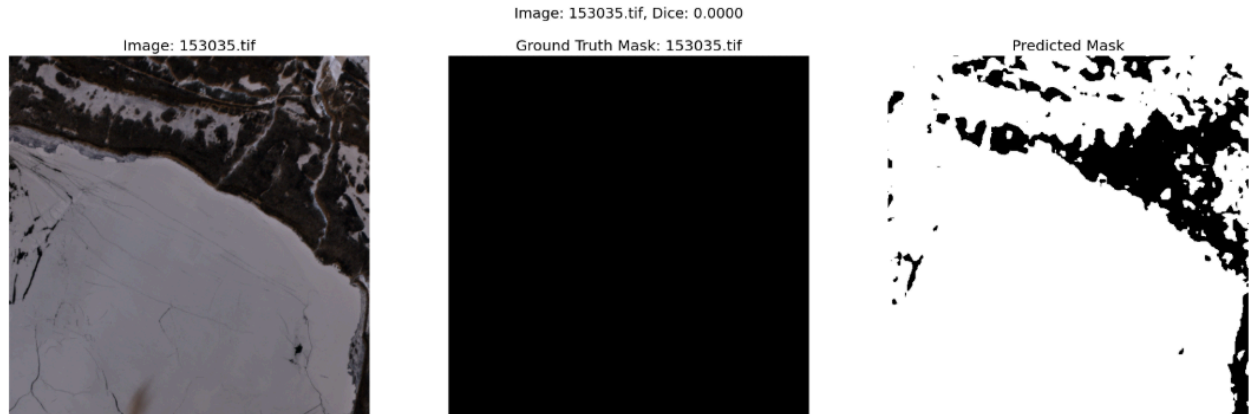
7. Performance Analysis Module

The primary metric used for evaluation during training and model selection was the Dice Coefficient, which measures the overlap between the predicted mask and the ground truth mask. Through the 5-fold cross-validation process described in Section 6, the DeeplabV3+ model demonstrated decent performance. The average Dice Coefficient achieved across the validation sets of all folds was approximately 0.93~0.94.

8. Enhancements and Future Work

While the developed DeeplabV3+ model achieved strong performance, several fields exist for further enhancement and future research to improve cloud masking accuracy and robustness, particularly addressing challenging scenarios identified during performance analysis.

- **Advanced Texture Analysis:** A key area for improvement lies in enhancing the model's ability to differentiate between clouds and spectrally similar bright surfaces, most notably snow and ice. As observed in the following figure, current CNN and Transformer models, while powerful feature extractors, can sometimes struggle to capture the subtle textural differences between these features, leading to misclassifications.



- **Ensemble Methods:** Combining predictions from multiple diverse models potentially using other valuable models like segformer or cloudformer which often leads to more robust and accurate results than any single model.

9. Conclusion

This project successfully created an AI system for cloud masking using DeeplabV3+, achieving a robust solution through 5-fold cross-validation and noise-aware training. The model performed well, scoring 0.8966 Dice on the noisy test set. Major challenges included the impact of label noise and differentiating clouds from features like snow, indicating a need for better texture analysis. Despite this, the project confirmed DeeplabV3+'s effectiveness, suggesting future work should focus on texture analysis and data cleaning.

10. Workload Distribution

Team Member	ID	Contribution(s)
Ahmed Osama Helmy	9213061	Segformer Model-Project Report
Aliaa Abelaziz	9210694	Segformer Model-Project Report
Abdallah Ahmed Ali	9210652	DCNet/U-Net-Profiler
Omar Mahmoud	9210758	Preprocessing/Augmentation DeeplabV3+ Profiler/Submission

11. References/Appendix

Model	Accuracy (Dice Coefficient)
U-Net	0.78~0.80 (Uncleaned Data)
U-Net++	~0.84
DeeplabV3+ (ResNet-50 Backbone)	0.8966 (0.94 on cleaned)
SegFormer (B2)	~0.84
SegFormer (B3)	~0.85
Cloudformer	~0.86

These values aren't for reference and can't be used to compare these models as some of them were done without preprocessing or cleaning or parameter tuning.

Any of these models could achieve the same accuracy as DeeplabV3+ or potentially exceed it if trained correctly.

Model Complexity Analysis

The complexity of the final DeeplabV3+ model was analyzed using a custom profiler. The results for a standard input size corresponding to the project's data (1 batch, 4 channels, 512x512 pixels) are as follows:

- **Input Size:** (1, 4, 512, 512)
- **Computational Operations:** 73.6944 GOps (Giga Operations)
- **Number of Trainable Parameters:** 26.6807 M (Million)

These values were obtained using our profiling tools and are consistent with the known parameter counts and computational load expected for a DeeplabV3+ architecture utilizing a ResNet-50 backbone, adapted for a 4-channel input.

1. [Semantic Segmentation of Remote Sensing Images Using Deep Learning: A Survey](#)
2. [Cloudformer: A Cloud-Removal Network Combining Self-Attention Mechanism and Convolution](#)
3. [SegFormer: Simple and Efficient Design for Semantic Segmentation with Transformers](#)
4. [U-Net: Convolutional Networks for Biomedical Image Segmentation](#)
5. [U-Net++: A Nested U-Net Architecture for Medical Image Segmentation](#)
6. [Encoder-Decoder with Atrous Separable Convolution for Semantic Image Segmentation](#)